

Data Storytelling

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Spring 2026

Week 13

Announcements

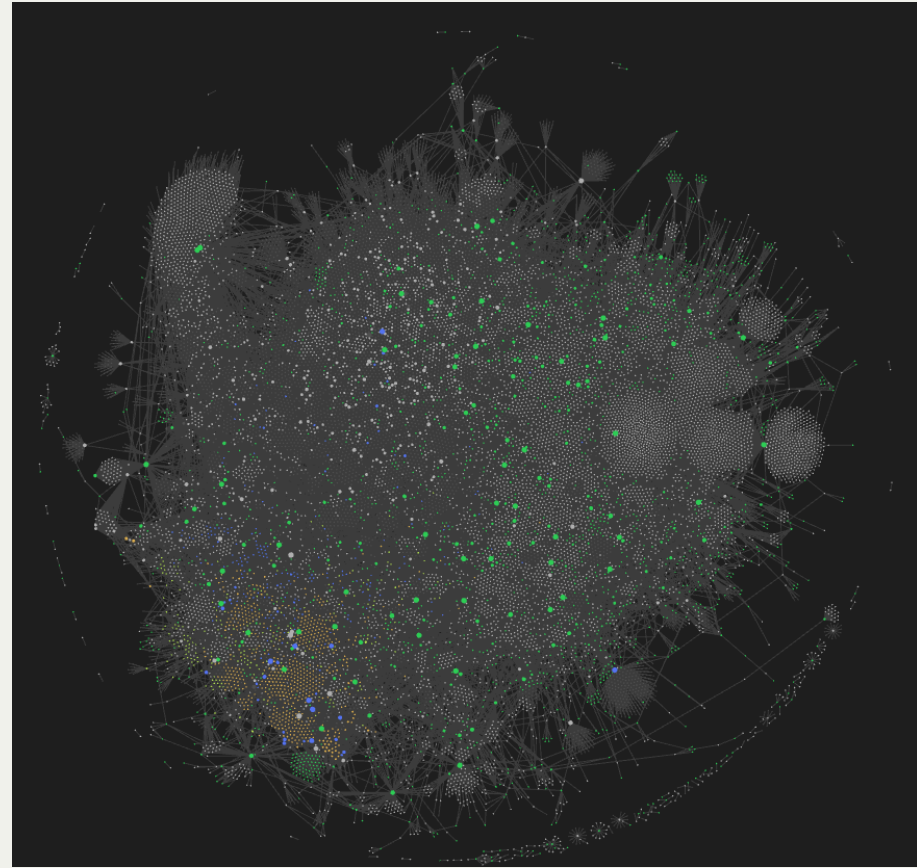
- Homework 2 Due tomorrow (Tuesday, 11:59 PM)
- Final paper due May 15th at 11:59 PM
- Final presentation

Network Analysis

What is a network?

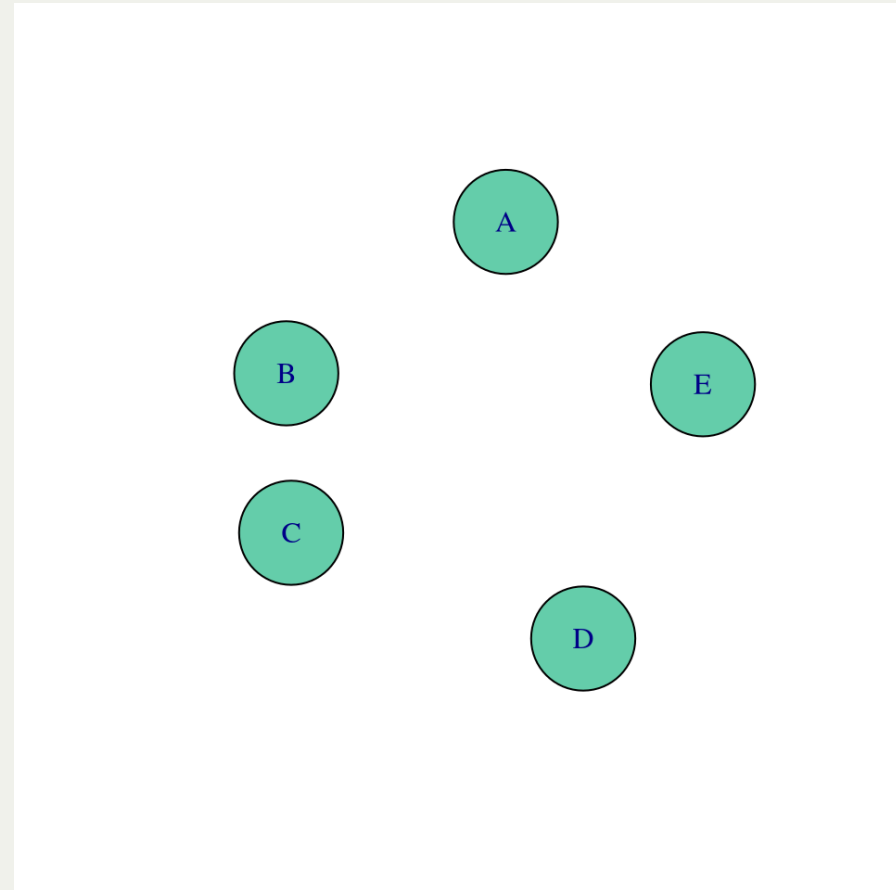
- **Networks** a structure made up of **nodes** (also called vertices) and **links** (also called edges) that connect pairs of nodes.
 - Used to represent relationships between both theoretical constructions and real-world entities.
 - *Theoretical*: e.g., a network of concepts in a theory, a network of arguments in a debate, etc.
 - *Real-world*: e.g., a network of people in a social group, a network

Obsidian Knowledge Graph



Nodes (Vertices)

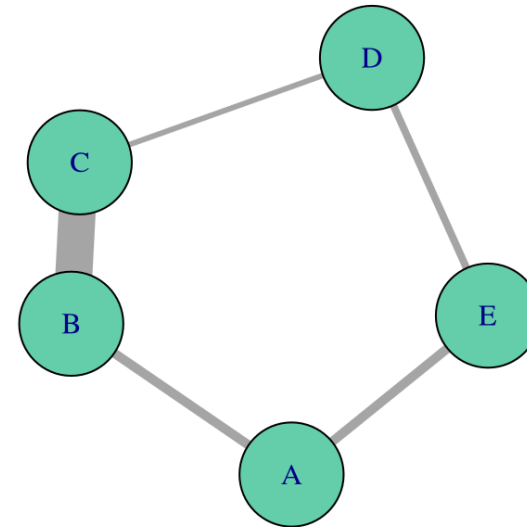
- **Nodes**
 - An individual entity in a network.
 - Can correspond to a person, organization, concept, etc.
- Nodes can have *attributes* or characteristics that may be relevant to the analysis.
 - Ex. gender, age, location, etc. for a person; size, industry, etc. for an organization; type, category, etc.



Links (Edges)

Links

- A connection between two nodes in a network
- Typically represented as a line connecting two nodes in a network diagram.
- Symbolize a relationship, interaction, or flow of information between the nodes.
- Links can be *directed* (e.g., $A \rightarrow B$) or *undirected* (e.g., $A - B$).
- Links can also have *weights* that indicate the strength or intensity of the relationship.



Examples of networks

- **Social networks**

Adjacency matrix

- An adjacency matrix is a square matrix that represents the connections between nodes in a network.
- The rows and columns of the matrix correspond to the nodes in the network, and the entries indicate the presence or absence of a link between the nodes.
- For an *undirected network*, the adjacency matrix is symmetric, while for a directed network, it may not be symmetric.

$$A = \begin{bmatrix} 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \end{bmatrix}$$

Network metrics

Centrality

Clustering

Sources of network data